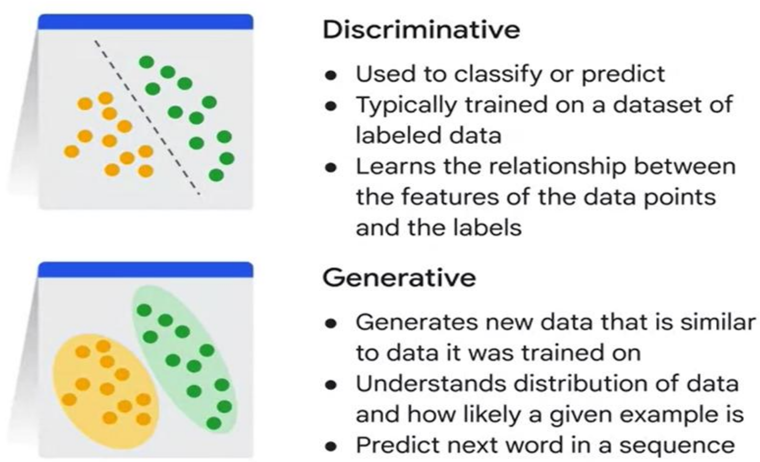
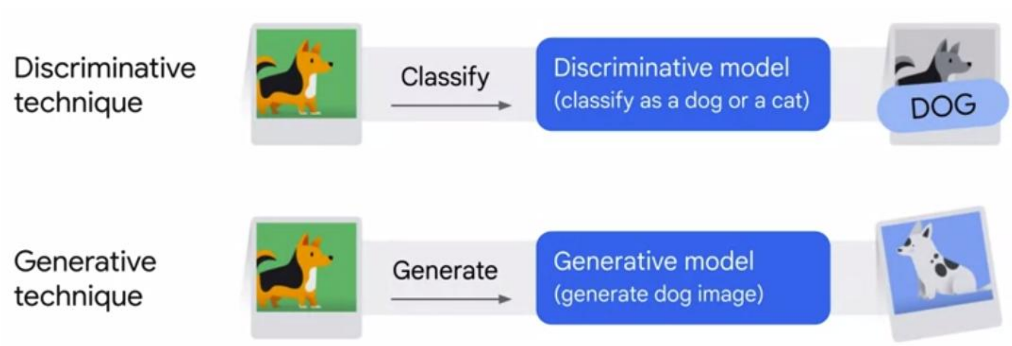
Deep Learning (CAI-466)

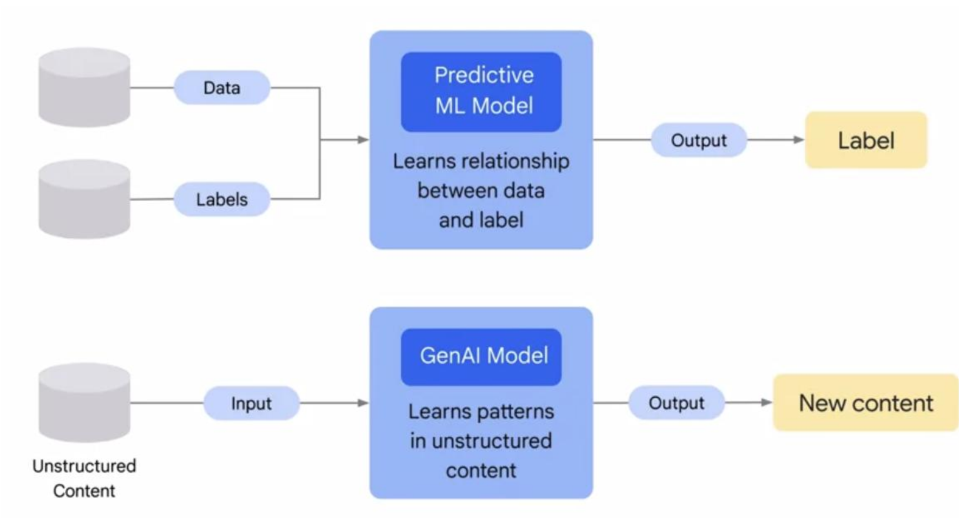
Objectives:

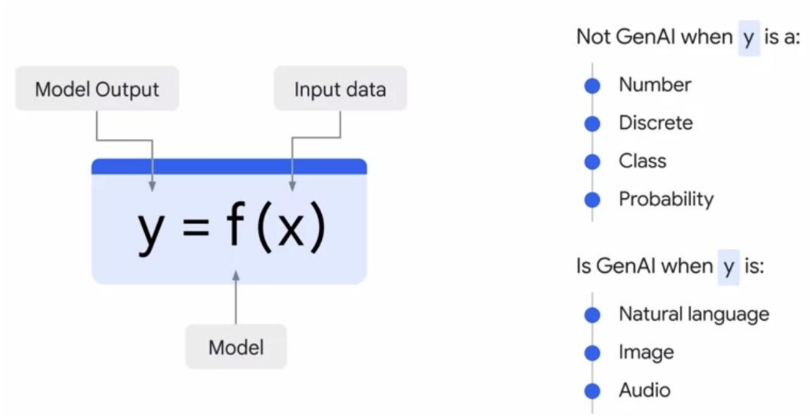
- **Types of AI**
- **Generative AI Model Types**
- **Large Language Models**

Types of AI







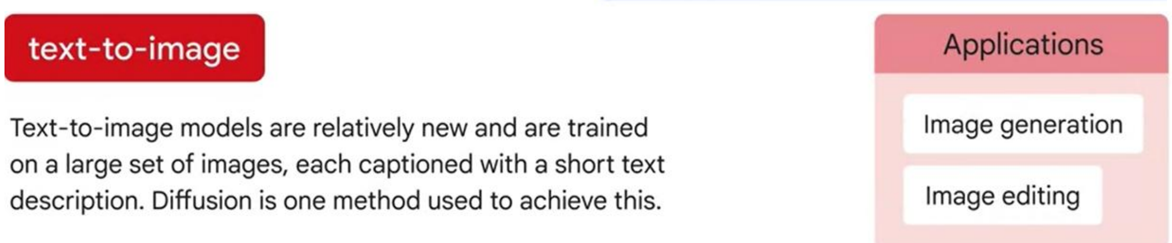


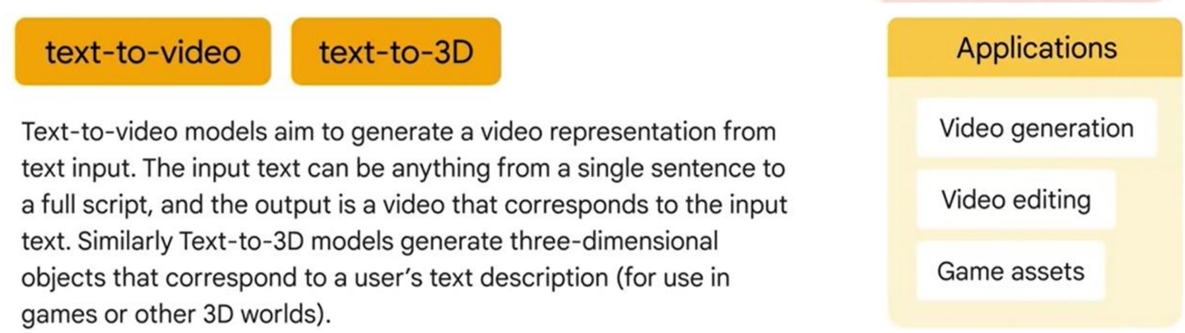
• GenAl is a type of Artificial Intelligence that creates new content based on what it has learned from existing content.

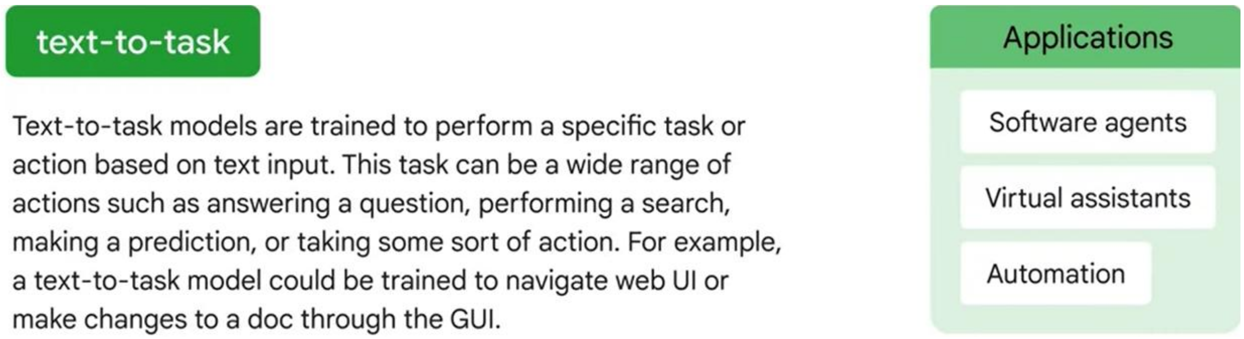
• The process of learning from existing content is called training and results in the creation of a statistical model.

• When given a prompt, GenAl uses this statistical model to predict what an expected response might be-and this generates new content.

Generative AI Model Types



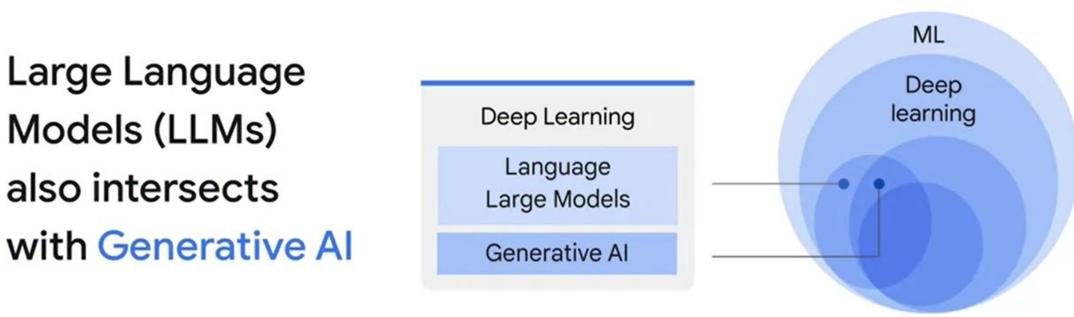


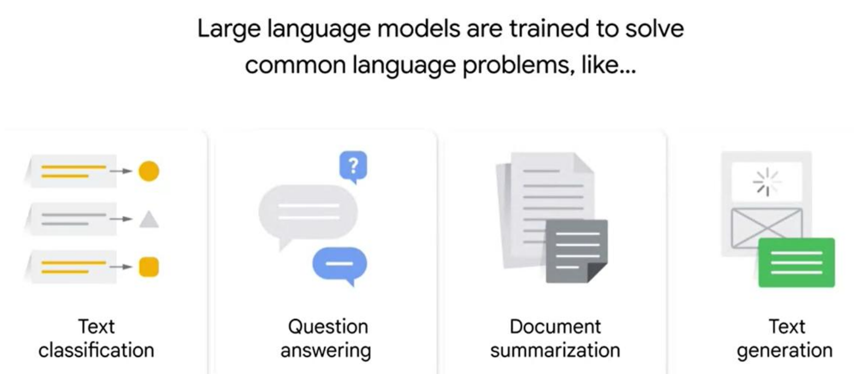


Large Language Models (LLMs)

Large language models, also known as LLMs, are very large deep learning models that are pre-trained on vast amounts of data. The underlying transformer is a set of neural networks that consist of an encoder and a decoder with self-attention capabilities. The encoder and decoder extract meanings from a sequence of text and understand the relationships between words and phrases in it.

Transformer LLMs are capable of unsupervised training, although a more precise explanation is that transformers perform self-learning. It is through this process that transformers learn to understand basic grammar, languages, and knowledge.





Examples of popular large language models

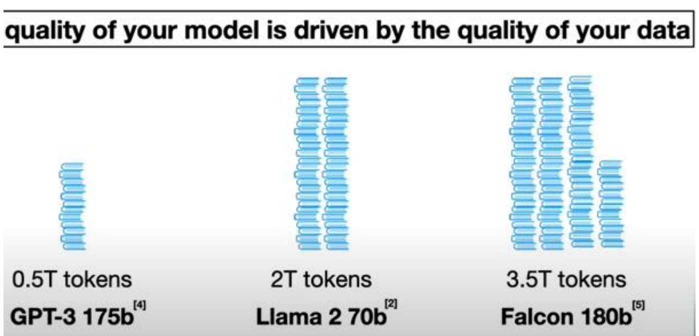
Popular large language models have taken the world by storm. Many have been adopted by people across industries. Some popular LLM models include:

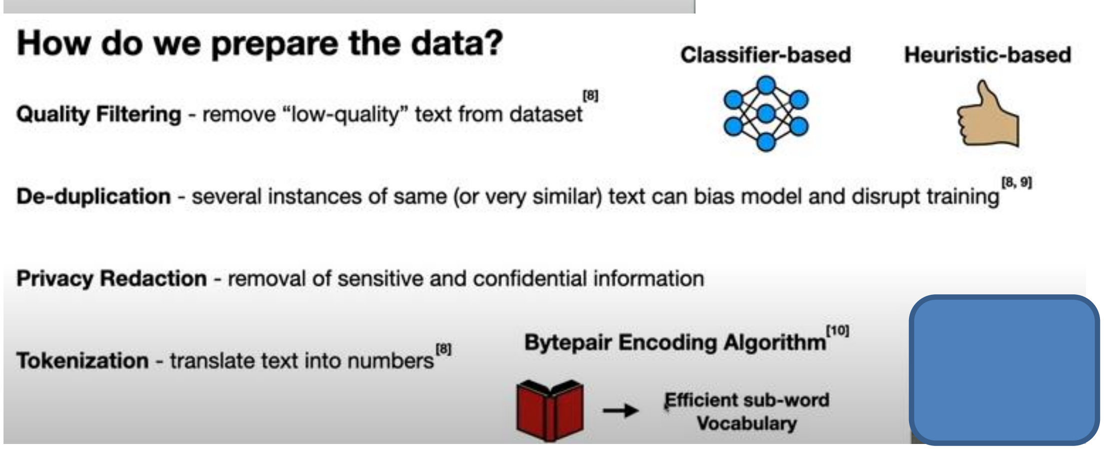
- **PaLM:** Google's Pathways Language Model (PaLM) is a transformer language model capable of common-sense and arithmetic reasoning, joke explanation, code generation, and translation.
- **BERT:** The Bidirectional Encoder Representations from Transformers (BERT) language model was also developed at Google. It is a transformer-based model that can understand natural language and answer questions.
- **XLNet:** A permutation language model, XLNet generated output predictions in a random order, which distinguishes it from BERT. It assesses the pattern of tokens encoded and then predicts tokens in random order, instead of a sequential order.
- **GPT:** Generative pre-trained transformers are perhaps the best-known large language models. Developed by OpenAI, GPT is a popular foundational model whose numbered iterations are improvements on their predecessors (GPT-3, GPT-4, etc.). It can be fine-tuned to perform specific tasks downstream. Examples of this are EinsteinGPT, developed by Salesforce for CRM, and Bloomberg's BloombergGPT for finance.

Steps required In LLM development

1) Data Curation
2) Model Architecture
3) Training at scale
4) Evaluation

Step 1: Data Curation





Where do we get all these data?

The internet e.g. web pages, wikipedia, forums, books, scientific articles, code bases, etc.

Public datasets
- Common Crawl (Colossal Clean Crawled Corpus i.e. C4, Falcon RefinedWeb)
- The Pile[6]
- Hugging Face Datasets

Private data sources e.g. FinPile (BloombergGPT) drawn from Bloomberg archives[1]

Use an LLM e.g. Alpaca - an LLM trained on structured text generated by GPT-3[7]

Step2: Model Architecture:

Transformers

Neural network architecture that uses **attention** mechanisms

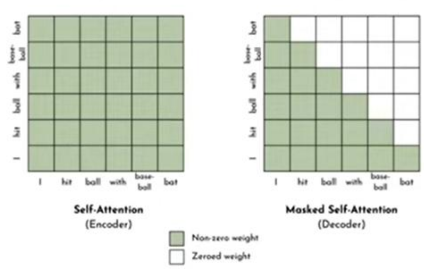
Attention mechanism - learns dependencies between different elements of a sequence based on position and content [13]

"I hit the baseball with a bat" *"I hit the bat with a baseball"*

3 Types of Transformers [14, 15]

Encoder-only - encoder translates tokens into a semantically meaningful representation | *tasks: text classification* [15]

Decoder-only - similar to encoder but does not allow self-attention with future elements | *tasks: text generation* [8, 15]



Encoder-decoder - combines both and allows cross-attention | *tasks: translation* [13, 15]

Step3: Training at Scale

Three common learning models exist:

- Zero-shot learning; Base LLMs can respond to a broad range of requests without explicit training, often through prompts, although answer accuracy varies.
- Few-shot learning: By providing a few relevant training examples, base model performance significantly improves in that specific area.
- Fine-tuning: This is an extension of few-shot learning in that data scientists train a base model to adjust its parameters with additional data relevant to the specific application.

3 Training Techniques

Mixed Precision Training - uses both 32-bit and 16-bit floating point data types[8, 22]

3D Parallelism - combination of pipeline, model, and data parallelism[8]

- **Pipeline Parallelism** - distributes transformer layers across multiple GPUs
- **Model Parallelism** - decomposes parameter matrix operation into multiple matrix multiplies distributed across multiple GPUs
- **Data Parallelism** - distributes training data across multiple GPUs

Zero Redundancy Optimizer (ZeRO) - reduces data redundancy regarding the optimizer state, gradient, or parameter partitioning[8]

Hyperparameters

Batch Size: (*Static*) typically ~16M tokens. (*Dynamic*) GPT-3 increased from 32K to 3.2M [8]

Learning Rate: (*Dynamic*) learning rate increases linearly until reaching a maximum value and then reduces via a cosine decay until the learning rate is about 10% of its max value[8]

Optimizer: Adam-based optimizers are most commonly used for LLMs[8]

Dropout: typical values between 0.2 and 0.5[32]

[2]

Llama 2 (7b)	~180,000 GPU hours	→	**~10b model**	~100,000 GPU hours
Llama 2 (70b)	~1,700,000 GPU hours		**~100b model**	~1,000,000 GPU hours

Renting

Invidia A100: $1-2 per GPU per hour

⟹ **10b model: $150,000**
100b model: $1,500,000

Buying

Invidia A100: ~$10,000

⟹ **GPU Cluster**: ~$10,000 x 1000 = **$10,000,000**

Training Energy Cost (100b model): ~1,000 megawatt hour[3]
Energy Price: ~$100 per megawatt hour

⟹ **Marginal Energy cost (100b model)**:
1,000 x $100 = **$100,000**

Step4: Evaluation

The Open LLM Leaderboard, hosted on Hugging Face, evaluates and ranks open-source Large Language Models (LLMs) and chatbots. It serves as a resource for the AI community, offering an up-to-date, benchmark comparison of various open-source LLMs. This platform enables the submission of models for automated evaluation, ensuring standardized and impartial assessment of each model's capabilities.

This leaderboard is part of The Big Benchmarks Collection on Hugging Face, which encompasses a broader set of benchmarks such as MTEB and LLM-Perf. These benchmarks help compare strengths and weaknesses of different models across varied tasks and scenarios.

The Open LLM Leaderboard ranks models on several benchmarks including ARC, HellaSwag and MMLU, and makes it possible to filter models by type, precision, architecture, and other options.